STRICT TRANSFORMS COMMUTE WITH PUSHFORWARD ALONG AN AFFINE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Affine base change of pushforwards. Cohomology of Schemes, Lemma coherent-lemma-affine-base-change, used at proof lines 8920-8922. Identifies g'_* of the pulled back sheaf.
- Pushforward of sections supported on a closed subset. Properties, Lemma properties-lemma-push-sections-supported-on-closed-subset, used at proof lines 8924-8928. Identifies the supported subsheaves.
- Base change of an affine morphism is affine. Morphisms, Lemma morphisms-lemma-base-change-affine, used at proof lines 8928-8931. Makes g'_* exact, which turns the identification of subsheaves into the claim.

Source reference. Original lemma 02KG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3371–3371. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01SD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2334–2334. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07ZQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10155–10155. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $Z \subset S$ be a closed subscheme. Let $b : S' \to S$ be the blowing up of Z in S. Let $g : X \to Y$ be an affine morphism of schemes over S. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $g' : X \times_S S' \to Y \times_S S'$ be the base change of g. Let $\mathcal{F}'$ be the strict transform of $\mathcal{F}$ relative to b. Then $g'_*\mathcal{F}'$ is the strict transform of $g_*\mathcal{F}$.*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Observe that $g'_*\mathrm{pr}_X^*\mathcal{F} = \mathrm{pr}_Y^* g_*\mathcal{F}$ by Cohomology of Schemes, Lemma 02KG. Let $\mathcal{K} \subset \mathrm{pr}_X^*\mathcal{F}$ be the subsheaf of sections supported in the inverse image of Z in $X \times_S S'$. By Properties, Lemma 07ZQ the pushforward $g'_*\mathcal{K}$ is the subsheaf of sections of $\mathrm{pr}_Y^* g_*\mathcal{F}$ supported in the inverse image of Z in $Y \times_S S'$. As g' is affine (Morphisms, Lemma 01SD) we see that g'_* is exact, hence we conclude. $\square$